

Department of Information Technology
B.Tech IV Year
Lecture Plan

Subject Code: IT48330
Computing

Session: July-Dec 2022

Subject Nomenclature: Cloud

Semester: “A”

Lecture No.	Unit No.	Topic Covered
1	I	Historical Development, Vision of Cloud Computing.
2	I	Characteristics of cloud computing as per NIST, Cloud services requirements.
3	I	Cloud and dynamic infrastructure, Components of cloud computing
4	I	Deployment models their working with real life example.
5	I	Services of cloud computing SAAS, PAAS, IAAS and Others
6	I	Organizational scenarios of clouds
7	I	Administering & Monitoring cloud services,
8	I	Benefits and limitations, Deploy application over cloud.
9	II	Cloud Reference Model, Types of Clouds
10	II	Cloud Interoperability & Standards
11	II	Scalability and Fault Tolerance feature of cloud
12	II	Cloud Solutions: Cloud Ecosystem
13	II	Cloud Business Process Management
14	II	Cloud Service Management, Micro Services & Monolithic architecture.
15	II	Cloud Offerings: Cloud Analytics, Testing Under Control, Virtual Desktop Infrastructure.
15	III	Resiliency, Provisioning, Asset management,
16	III	Concepts of Map-reduce, Cloud Governance
17	III	Introduction to Hadoop
18	III	High Availability and Disaster Recovery.
19	III	Different Data Recovery technique
20	III	Virtualization: Fundamental concepts of Compute
21	III	Storage, Networking, desktop, and application virtualization.
22	III	Virtualization benefits, server virtualization
23	III	Block and file-level storage virtualization
24	III	Types of Hypervisor & Hypervisor management software,
25	III	Infrastructure Requirements, Virtual LAN(VLAN), and Virtual SAN (VSAN) and their benefits.
26	IV	Need of Cloud Information security & its fundamentals
27	IV	Cloud security services with a Microsoft Azure service
28	IV	Design principles, Secure Cloud Software Requirements,
29	IV	Policy Implementation for cloud security

30	IV	Cloud Computing Security Challenges.
31	IV	Virtualization Security Management, Cloud Computing Security Architecture.
32	V	Characterization & Definition, Cloud Federation Stack
33	V	Third-Party Cloud Services. Case study: Google App Engine
34	V	Google Cloud Service
36	V	Microsoft Azure
38	V	AWS
40	V	Discuss different between Azure, GCP, and AWS.
